

Poisson process

A process $N(t)$ indexed by $t \in [0, \infty)$ will be called a counting pr.

if

$$N(t) \geq 0, \quad N(t) \in \{0, 1, 2, \dots\}.$$

$$N(t) \geq N(s) \quad \text{if } t > s.$$

Increment $\rightarrow N(t+s) - N(s)$ is called
the increment for period $(s, t+s]$.

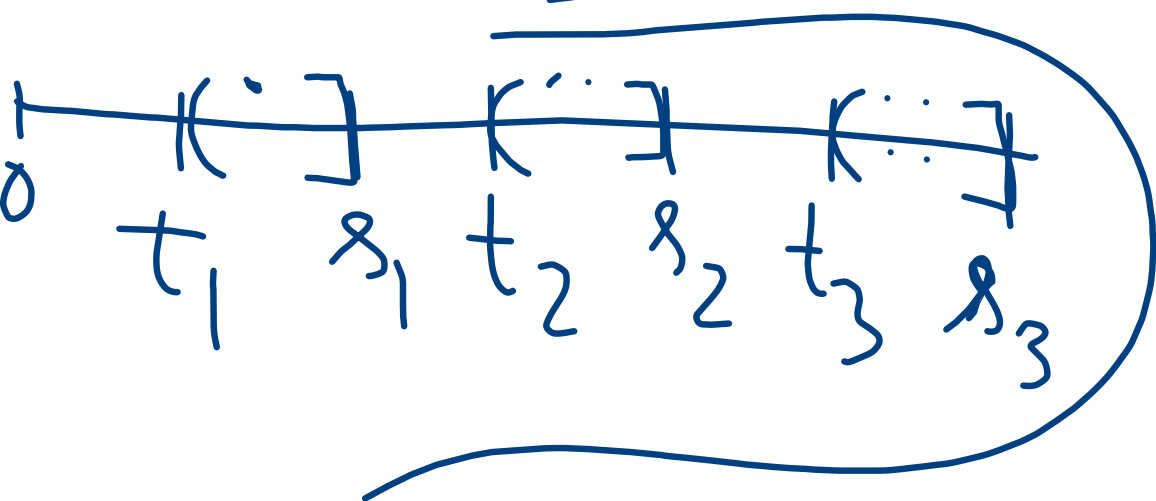
• Increments are stationary if the
distrⁿ of $N(t+s) - N(s)$ does not
depend on s , and only on t .

• Inc. are indept. if for any $n \geq 1$,
 $0 \leq t_1 < s_1 \leq t_2 < s_2 \leq t_3 < \dots \leq t_n < s_n$.

we have,

$\{N(s_1) - N(t_1), N(s_2) - N(t_2), \dots, N(s_n) - N(t_n)\}$
is an indept. family.

Defn:- A counting process is called



a homogeneous Poisson pr.
with intensity $\lambda > 0$, if

- $N(0) = 0$ w.p. 1.
- N has indept inc.

$$P(N(t+s) - N(s) = k) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}.$$

for $0 \leq s, t > 0$.

Defⁿ: A fn $f: \mathbb{R} \rightarrow \mathbb{R}$ is called

$o(h)$ if

$$\lim_{h \rightarrow 0} \frac{f(h)}{h} = 0.$$

$$f(x) = x^2,$$

$$f(h)/h = h \rightarrow 0.$$

$$f = o(h)$$

$$f_1 = o(h), f_2 = o(h),$$

$$\frac{f_1(h) + f_2(h)}{h} \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

$$\Rightarrow f_1 + f_2 = o(h).$$

$$P(N(t+h) - N(t) = 1) = e^{-\lambda h} \cdot \lambda h$$

i.e.,

$$\frac{e^{-\lambda h} \lambda h - \lambda h}{h} = \frac{\lambda h}{h} (e^{-\lambda h} - 1) \rightarrow 0 \quad \text{--- } \textcircled{x}$$

$$P(N(t+h) - N(t) \geq 2) = o(h) \quad \text{---} \quad \textcircled{\ast \ast}$$

$$1 - e^{-\lambda h} - \lambda h e^{-\lambda h}$$

$$\frac{P(N(t+h) - N(t) \geq 2)}{h} = \frac{1 - e^{-\lambda h} - \lambda h e^{-\lambda h}}{h}$$

$$\frac{P(N(t+h) - N(t) = 0)}{h}$$

$$P(N(t+h) - N(t) = 0) \rightarrow 0 \quad \text{use L'Hospital.}$$

$$= 1 - P(N(t+h) - N(t) = 1) - P(N(t+h) - N(t) \geq 2)$$

$$= 1 - \lambda h + o(h)$$

Thm A counting pr: with indept.
increments and satisfying

• $N(0) = 0$ w.p. 1.

• $(*)$ & $(**)$ are satisfied.

Then it must be a Poisson pr.

Pf:- We need to show:-

$$N(t+s) - N(s) \sim \text{Poi}(\lambda t).$$

$P_{nt} \text{ } s=0$. define $P_k(t) = P(N(t)=k)$.

For $k=0$, $P_0(0) = P(N(0)=0)$

$$P_0(t+h) = P(N(t+h)=0)$$

$$= P(N(t)=0, N(t+h)=0)$$

$$= P(N(t)=0, N(t+h)-N(t)=0)$$

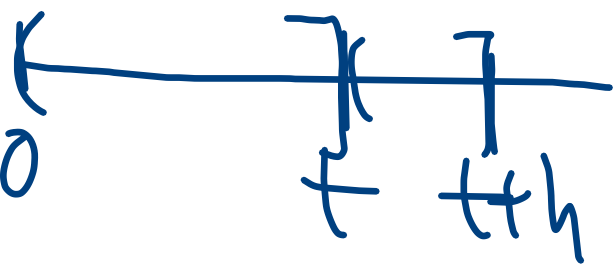
$$= P(N(t)=0) P(N(t+h)-N(t)=0)$$

$$= P_0(t) (1 - \lambda h + o(h))$$

$$\left(= \frac{e^{-\lambda t} (\lambda t)^k}{k!} \right)$$

$$\frac{P_0(t+h) - P_0(t)}{h}$$

$$= -\frac{\lambda h P_0(t)}{h} + \frac{o(h) P_0(t)}{h}$$



Taking limit as $h \rightarrow 0$,

$$P_0'(t) = -\lambda P_0(t),$$

$$P_0'(t) e^{\lambda t} + \lambda e^{\lambda t} P_0(t) = 0$$

$$\Rightarrow e^{\lambda t} P_0(t) = \text{Const.}$$

$$\Rightarrow \frac{d}{dt} (e^{\lambda t} P_0(t)) = 0$$

Putting $t = \infty$, $1 = \text{Const.}$

$$\Rightarrow P_0(t) = e^{-\lambda t}.$$

Suppose the result holds for k .

$$P_k(t) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}.$$

$$\begin{aligned}
 P_{k+1}(t+h) &= P(N(t+h) = k+1) \\
 &= P(N(t+h) = k+1, N(t) = k+1) \\
 &\quad + P(N(t+h) = k+1, N(t) = k) \\
 &\quad + \sum_{j=0}^{k-1} P(N(t+h) = k+1, N(t) = j)
 \end{aligned}$$

$$\begin{aligned} & P(N(t+h) = k+1, \\ & \quad N(t) = j) \\ &= P(N(t) = j, \\ & \quad N(t+h) - N(t) \\ & \quad \quad = k+1-j) \\ &\leq P(N(t+h) - N(t) \\ & \quad \quad \geq 2) \\ &= o(h) \end{aligned}$$

for $j \geq 0, 1, \dots, k-1$

$$\begin{aligned}
 & P(N(t+h)=k+1, N(t)=k) \\
 & \Rightarrow P(N(t)=k) P(N(t+h)-N(t)=1) \\
 & = P_k(t) (\lambda h + o(h)).
 \end{aligned}$$

$$P(N(t+h)=k+1, N(t)=k+1).$$

$$\begin{aligned}
 & = P(N(t)=k+1) P(N(t+h)-N(t)=0) \\
 & = P_{k+1}(t) (1 - \lambda h + o(h)).
 \end{aligned}$$

$$P_{k+1}(t+h) = P_{k+1}(t)(1 - \lambda h + o(h)) \\ + P_k(t)(\lambda h + o(h)) + o(h).$$

$$\Rightarrow \frac{P_{k+1}(t+h) - P_{k+1}(t)}{h} = -\frac{\lambda h P_{k+1}(t)}{h} + \frac{\lambda h P_k(t)}{h} + \frac{o(h)}{h}$$

Letting $h \rightarrow 0$,

$$P'_{k+1}(t) = -\lambda P_{k+1}(t) + \lambda P_k(t)$$

$f \geq 0$.

$f \leq g = o(h)$.

$\left| \frac{f(h)}{h} \right| \leq \frac{g(h)}{h} \rightarrow 0.$

$$P'_{k+1}(t) e^{\lambda t} + \lambda e^{\lambda t} P_{k+1}(t) = \lambda e^{\lambda t} P_k(t).$$

$$\Rightarrow \frac{d}{dt} (e^{\lambda t} P_{k+1}(t)) = \lambda e^{\lambda t} P_k(t)$$

$$\int_0^s \frac{d}{dt} () dt = \int_0^s \lambda \cancel{e^{\lambda t}} \frac{\cancel{e^{-\lambda t}} (\lambda t)^k}{k!} dt$$

$$\Rightarrow e^{\lambda s} P_{k+1}(s) - \cancel{e^{\lambda 0}} \cdot P_{k+1}(0) = \frac{\lambda^{k+1}}{k!} \int_0^s t^k dt = \frac{\lambda^{k+1}}{k! (k+1)} s^{k+1}$$

$$\Rightarrow P_{k+1}(t) = e^{-\lambda t} \frac{(\lambda t)^{k+1}}{(k+1)!}$$

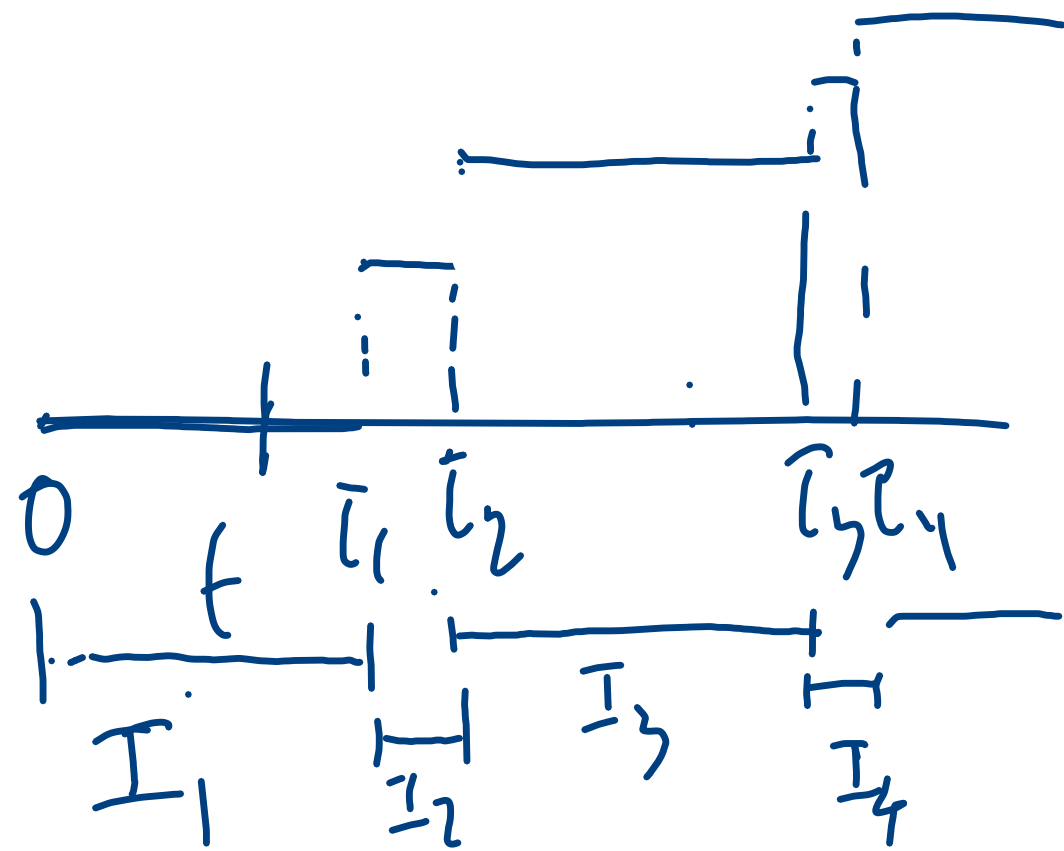
$$P_{k+1}(0)$$

$$= P(N(0) = k+1).$$

Arrival times :-

$$\tau_1 = \inf \{t : N(t) = 1\}$$

$$\tau_2 = \inf \{t : N(t) = 2\},$$



In gen. $k \geq 1$,

$$\tau_k = \inf \{t : N(t) = k\}$$

Set $\tau_0 = 0$.

$$P(\tau_1 > t) = P(N(t)=0).$$

$$\stackrel{I_1}{=} e^{-\lambda t}.$$

Hence, $\tau_1 \sim \text{Exp}(\lambda).$

Define inter. arrival times,

$$I_k = \tau_k - \tau_{k-1}, \quad k \geq 1.$$

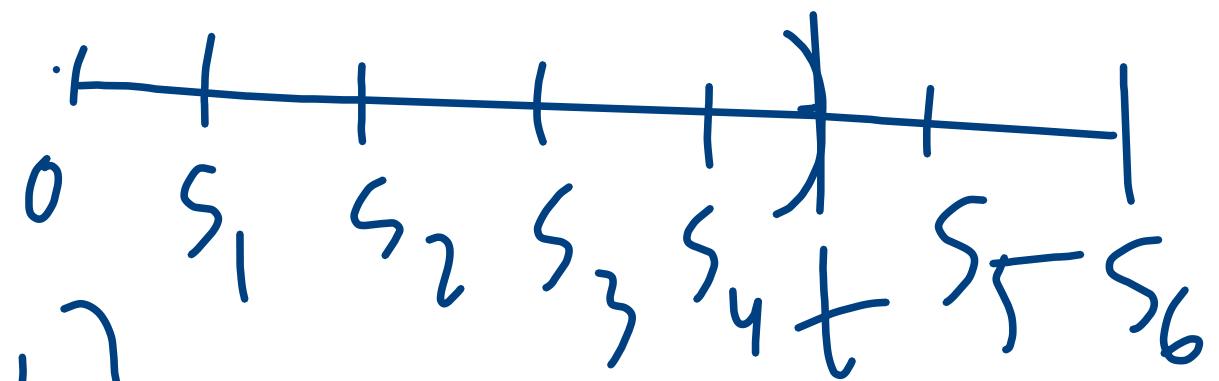
$$\begin{aligned} \tau_2 &= \tau_2 - \tau_1 + \tau_1 \\ &= I_2 + I_1. \end{aligned}$$

$$\begin{aligned} &P(\tau_2 > t) \\ &= P(\tau_1 > t) \\ &\quad + P(\tau_1 \leq t, \tau_2 > t) \\ &= P(N(t)=0) \\ &\quad + P(N(t)=1) \\ &= e^{-\lambda t} (1 + \lambda t) \end{aligned}$$

Propⁿ :- The inter arrival times are iid.
exp.

$E_1, E_2, \dots$ iid $\exp(\lambda)$.

$$S_n = \sum_{i=1}^n E_i$$



$$N(t) = \max\{n : S_n \leq t\}$$

$\sim \text{Poi pr.}$

$$e^{-100} \quad E_1 \sim \exp(1)$$